# Design Pattern Identification and Evolution

## in a Ruby on Rails-Based Travel Planning Application

Case Study Application: DaPlanStan

Team B5 | Batch CSE B | 23CSE455 Design Patterns

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Pattern discovery • software evolution • cross-language comparison • LLM records • human verification

Final case-study summary

# Application and architecture

DaPlanStan is a self-hosted Rails travel-planning application.




### Core travel flows

Trip creation and membership
Itinerary and trip-item management
Booking-email ingestion and parsing

### Integrations

Anthropic for parsing / suggestions
Geocoding + maps
Action Mailer / Postmark

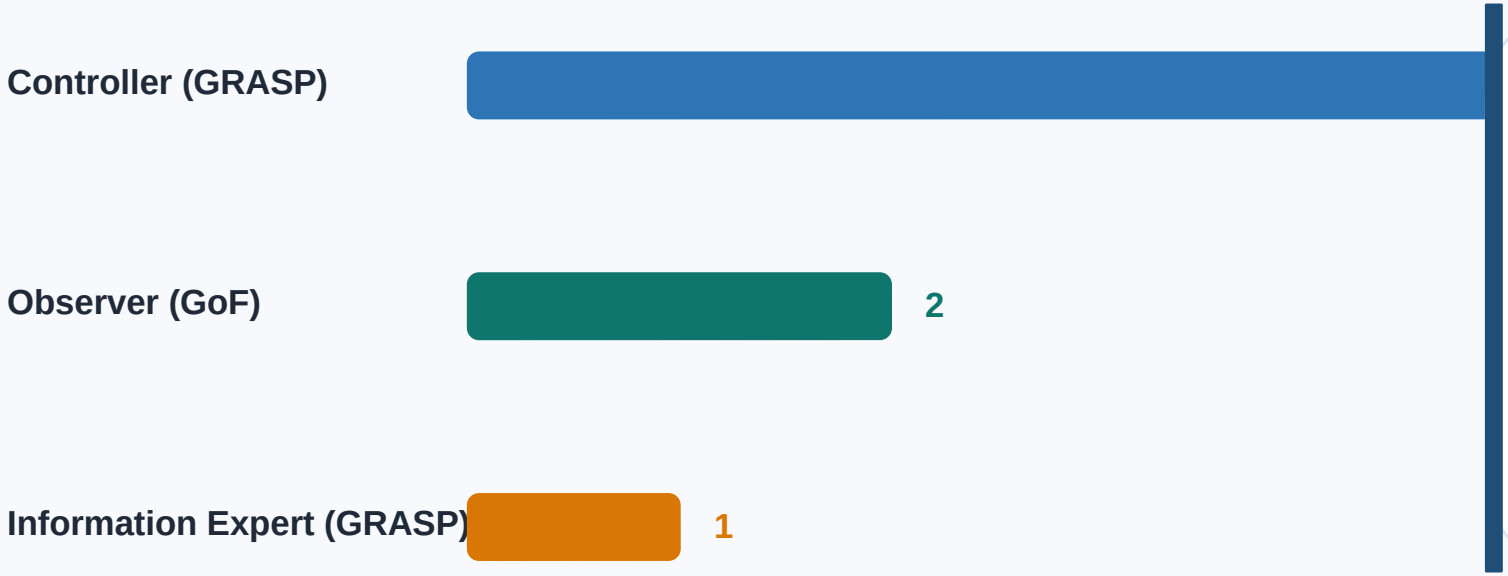
### Technology

Ruby 3.4.2 • Rails 8.1.3
SQLite • Hotwire/Turbo/Stimulus
Active Job and Action Mailbox

*Baseline source snapshot was inspected statically; exact ZIP-to-Git-commit identity remained unverified.*

# Part 1 - Pattern discovery

Frozen baseline: 9 confirmed pattern collaborations across 3 pattern types.




### Discovery method

Inspected application source and framework-mediated collaborations.

Mapped actual participants and file locations.

Rejected weak similarities and framework conventions that did not establish a full pattern collaboration.

## Confirmed baseline inventory

P01 Information Expert • P02-P07 Controller • P08-P09 Observer

Rejected candidates included geocoding Template Method / Strategy / Adapter, timeline-map Observer, and Beta Invite Information Expert.

# Baseline pattern inventory

Nine source-supported collaborations were frozen before evolution.

| ID | Pattern | Collaboration |
|---|---|---|
| P01 | Information Expert | Parsed Email to Trip Record Linking |
| P02 | Controller | Inbound Travel Email Intake |
| P03 | Controller | Trip Creation |
| P04 | Controller | Trip Membership Management |
| P05 | Controller | Bulk Itinerary Import |
| P06 | Controller | OAuth Sign-in / Identity Linking |
| P07 | Controller | Traveler Enrollment |
| P08 | Observer | Outbound Mail Delivery Logging |
| P09 | Observer | Trip-Scoped Turbo Refresh |

# Part 2 - Software evolution

Three realistic changes were implemented cumulatively from the frozen baseline.

### Change 1 - Personalized itinerary

New preference-driven itinerary generation workflow using Anthropic.

Owner-only preview and confirm flow.

Added one Controller collaboration.

Standalone verification: 6/6 cases, 20 assertions.

### Change 2 - Booking providers

Two differently shaped demo providers normalized through an Adapter.

Provider option coordination added through a Controller.

No live booking or payment.

Standalone verification: 10/10 cases, 67 assertions.

### Change 3 - Cancellation policies

Flexible, Moderate and Strict cancellation behavior encapsulated using Strategy.

Quote-only; no actual cancellation/refund.

Added cancellation coordination Controller.

Standalone verification: 10/10 cases, 45 assertions.

**Final evolved inventory: 14 confirmed instances across 5 pattern types.**

# Evolution impact on pattern inventory

The original nine collaborations were preserved or modified; five new collaborations were introduced.



# Part 3 - Cross-language comparison

One representative instance per detected pattern type was implemented in four languages.

| Pattern type | Java | Python | JavaScript | C++ |
|---|---|---|---|---|
| Controller | PASS | PASS | PASS | PASS |
| Information Expert | PASS | PASS | PASS | PASS |
| Observer | PASS | PASS | PASS | PASS |
| Adapter | PASS | PASS | PASS | PASS |
| Strategy | PASS | PASS | PASS | PASS |

**20 implementations**

5 pattern types × 4 languages

**192 / 192 tests**

All recorded fixture tests passed

**Measured consistently**

Warmup + five measured runs; median process time

# Runtime and verification evidence

Final Rails suite and application boot were verified after resolving environment and compatibility issues.

| 59 | 359 | 0 | 0 |
|---|---|---|---|
| test runs | assertions | failures | errors |

### Runtime fixes discovered by full Rails testing

- Booking-provider duplicate detection changed to an exact marker search.
- Two Minitest constructor stubs were made keyword-argument compatible.
- ActiveJob test adapter used during verification because the test DB lacked Solid Queue job tables.

### Application boot verification

Rails 8.1.3 started successfully in test environment under Puma with Ruby 3.4.2.

No live Anthropic request or manual end-to-end UI demonstration was claimed.

Evidence is retained as raw logs and verification notes in the team folder.

# LLM records and human verification

AI-assisted analysis was documented with provenance gaps stated explicitly rather than fabricated.

### LLM provenance audit

43 interactions examined
29 authentic file-backed prompt/response records
12 partial records
1 unrecoverable record
1 newly documented record

### Human verification

148 checklist items
143 verified
3 partially verified
2 cannot confirm

Reviewer distribution was team-reported.

### Authentic chat evidence

Architecture chat PDF
Pattern discovery chat PDF
Evolution chat PDF
Cross-language chat PDF
Final verification / documentation chat PDF

## Known limitations retained in the documentation

- Exact ZIP-to-baseline-commit identity cannot be independently proven from the archive alone.
- Historical model/date provenance is incomplete for some LLM interactions.
- Scenario 2 uses deterministic demo providers, not live booking APIs.
- Scenario 3 is cancellation-quote logic only; no real refund/cancellation transaction.

# Findings and team contributions

The case study demonstrates source-grounded pattern identification and pattern evolution under realistic requirements.

## Key findings

- Baseline: 9 instances / 3 pattern types.
- Evolved: 14 instances / 5 pattern types.
- New requirements naturally introduced Adapter and Strategy collaborations.
- Cross-language implementations preserved pattern responsibilities across four ecosystems.
- Verification exposed real integration issues that static inspection alone did not reveal.

## Team contribution map

Shreya E R - application understanding and architecture
S Hemarshitha - pattern discovery and verification
Aakash Choudhary - software evolution and refactoring
Manish S D A - cross-language analysis
M Tanmayi - LLM audit, human verification and final documentation

**Final result: a traceable evidence-backed study from original architecture → pattern discovery → evolution → cross-language comparison → human verification.**

# Thank you